Control-Variability Grid Analysis of Discrete and Impulsive ZMPC On Application to T1DM

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***Abstract*—Type 1 diabetes mellitus is one of the most common chronic and auto immune diseases worldwide and Colombia with a growth rate of 3% - 5% Annual. Current research in the treatment of this illness is aimed at the implementation of the Artificial Pancreas, a device that allows the regulation of blood glucose levels automatically. Recently, several model predictive controllers has been developed to accomplish that, in this work a performance comparison between standard zone model predictive control and a novel impulsive zone model predictive control scheme is presented for the glycemia regulation problem in type I diabetic patients. Both control schemes are developed in a new T1DM model with long-term scenario and realistic stability properties. The standard ZMPC with discrete input was proposed for the diabetes disease, this control scheme is compared with a impulsive ZMPC with artificial variables. The comparison is based on a control-variability grid analysis, the results show that the impulsive ZMPC has a better performance.**

I. Introduction

Type 1 diabetes mellitus (T1DM) is chronic and autoimmune disease that cause progressive destruction of the pancreatic β-cells. As a consequence, the natural endogenous production of insulin disappears, thus results in a dysfunctional glycemic regulation. In Latin America is estimated that the prevalence of T1DM will increase by 250% over the next 20 years and at present, its cost in Health budget is around 6%. The inclusion of insulin pump therapy had a good acceptance due to the positive clinical results on diabetes control, but its performance in Colombia and Latin America are unknown, especially in outpatient routine [1].

The idea of an artificial pancreas was visioned more than 50 years ago, a device that closes the loop between the measurements of continuous glucose monitoring system (CGM) and insulin pump injection/infusion for patients with T1DM. Nowadays, model predictive control (MPC) has been the most developed strategy for the artificial pancreas [2]–[6]. Zone-MPC is a type of model predictive control algorithm that calculates the optimal future inputs to maintain the predicted output within a desired zone rather than a single set-point [7], this can be useful to maintain normoglycemia levels, due of this standard zone MPC is the version most proposed to closed the loop in the artificial pancreas. On the other hand the ZMPC with artificial variables use intermediary variables that forced to lie in the equilibrium space, these equilibriums are characterized base on two underlying discrete time systems, then this kind of controller has an enlarged domain of attraction, depending on the controllable set to the entire equilibrium space, instead of the controllable set to a given point or invariant set.

in [8] is proposed a standard ZMPC for patients with T1DM, in this paper is used this control technique and the iZMPC with artificial variables running on a new T1DM model. This model is based on the insulin therapy, this practicality makes the model more approximate to reality with respect to others and gives the doctor the fundamental tools of insulin therapy, this model is supported by identification results from clinical data; moreover, can be simulated the glucose dynamics from long time (Simulation time ¿ two days) with realistic stability properties. Hence, in this paper is compared the standart ZMPC with discrete input [8] applied to the diabetes disease with the ZMPC with state window target [14] which has the following characteristics i) impulsive input due to the nature of the problem ii) artificial variables, give an enlarged domain of attraction, This allows a shorter prediction horizon, reducing the computational cost.

For comparison is used the control-variability grid analysis (CVGA) [6] for measuring the quality of this closed-loop glucose control for a cut of patients taken from [11] this method is for the visualization of the extreme glucose excursions caused by a control algorithm in a group of subjects, with each subject presented by one data point for any given observation period. This analysis shows that the iZMPC controller has a better performance than the standard ZMPC because that it presents less hyper- and hypoglycemia episodes with less time outside to the normoglycemia levels. Furthermore, it has a less pronounced peaks due to the meals perturbations.

This paper is organized as follows: Section 2 presents some useful preliminaries about impulsive linear systems. In Section 2.1 the selected T1DM patient model is introduced [11] and the sections 2.2 and 2.3 describe the discrete and impulsive scheme. Section 3 describes the zone MPC formulation for both strategies. Numerical results are presented in Section

4. Finally, the last section is devoted to conclusions and perspectives.

II. Preliminaries

The class of dynamic systems of interest in this paper consists in a set of linear impulsive first-order differential equations of the form

$$\begin{cases} \dot{x}(t) &= A_c x(t), \quad x(0) = x_0, \quad t \neq \tau_k, \\ x(\tau_k^+) &= x(\tau_k) + Bu(\tau_k), \qquad k \in \mathbb{N}, \end{cases} \tag{1}$$

where the independent variable $t \in \mathbb{R}$ denotes time, τ_k, $k \in \mathbb{N}$, denotes the impulse time instants, τ_k^+ denotes the time instant after τ_k, $x \in \mathcal{X} \subseteq \mathbb{R}^n$ denotes the (constrained) state vector and $u \in \mathcal{U} \subseteq \mathbb{R}^m$ denotes the (constrained) impulsive control inputs. Both constraint sets, $\mathcal{X}$ and $\mathcal{U}$, are compact sets and contain the origin in their interior. $A_c \in \mathbb{R}^{n\times n}$ is the continuous transition matrices, while $B \in \mathbb{R}^{n\times m}$ is the impulsive input matrix. Furthermore, as part of the system description, a target state set $\mathcal{X}^{Tar} \subset \mathcal{X}$ is defined, where the system is desired to be steered and kept.

Let $\kappa(\cdot)$ be a control law, in such a way that $u(\tau_k) = \kappa(x(\tau_k))$, for $k \in \mathbb{N}$. Then, the closed-loop impulsive system is described by:

$$\begin{cases} \dot{x}(t) &= A_c x(t), \quad x(0) = x_0, \quad t \neq \tau_k, \\ x(\tau_k^+) &= x(\tau_k) + B\kappa(x(\tau_k)), \quad k \in \mathbb{N}, \end{cases} \tag{2}$$

This way, the closed-loop trajectory is denoted by $x(t) = \phi_{cl}(t; x_0, \kappa(\cdot))$, for $t \geq 0$, with $\phi_{cl}(0; x_0, \kappa(\cdot)) = x_0$, and clearly, the jump depends now only on the state.

A. Type 1 diabetes model

According to [11], we consider the following affine continuous time model:

$$\dot{x}(t) = Ax(t) + B_u u(t) + B_r r(t) + E, \quad x(0) = x_0, \tag{3}$$

where $x(t) = [G(t) \quad I(t) \quad \dot{I}(t) \quad D(t) \quad \dot{D}(t)]'$, being G the glycemia (mg/dl), I the plasma insulin (insulinemia) (U/l) and D the digestion of CHO (g/dl). Furthermore, $u(t)$ is the insulin infusion rate (U) and $r(t)$ is the meal delivery rate (g/min). E is a constant term denoting the difference between the liver endogenous glucose production k_1 and the glucose absorption rate by the brain k_b.

The model matrices are given by:

$$A = \begin{pmatrix} 0 & -k_{si} & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & -\frac{1}{T_u^2} & -\frac{2}{T_u} & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & -\frac{1}{T_r^2} & -\frac{2}{T_r} \end{pmatrix}, \tag{4}$$

$$B_u = \begin{pmatrix} 0 \\ 0 \\ \frac{k_u}{V_i T_u^2} \\ 0 \\ 0 \end{pmatrix}, B_r = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ \frac{k_r}{V_B T_r^2} \end{pmatrix}, E = \begin{pmatrix} \theta \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix},$$

where k_{si} represents the sensitivity to insulin, T_u and $\frac{k_u}{V_i}$ are respectively the time constant and static gain of the relationship between the input $u(t)$ (insulin rate) and the insulinemia $I(t)$, with V_i being the insulin distributed volume, and T_r and $\frac{k_r}{V_B}$ are respectively the time constant and static gain of the relationship between the CHO in meal $r(t)$ and the digestion of CHO, $D(t)$, with V_B being the blood volume. Furthermore, the parameter θ of the constant term is given by $\theta = k_1 - k_b$.

Constraints for both, states and inputs are considered, in such a way that $u \in \mathcal{U}$, $x \in \mathcal{X}$ and $r \in \mathcal{R}$, where $\mathcal{U}$, $\mathcal{X}$ and $\mathcal{R}$ are assumed to be polyhedrons.

The main advantages of this model is that it is an integrating model (A has an eigenvalue at zero, which means that it is not stable) and so it better represents the evolution of the variable in real life diabetic patients. For instance, any equilibrium point corresponding to the fasting state ($r_s = 0$) is given by an arbitrary value of G, fixed values for the other states, and a fixed (and unique) value of insulin, $u_s \triangleq u_b$, which is known as the basal insulin rate. These characteristics, that make the model essentially different from other models used in the literature (mainly the Bergman model, [12]), allow us to make long term predictions. This way, it is argued that every undesirable episode can be better anticipated and corrected.

The general control objective is to maintain the glycemia in a safety range, $\mathcal{X}^{Tar}$, while keeping the other variables fulfilling the constraints, which are mainly positive values. An interesting point is that the hypoglycemia episodes (when glycemia is below this safety range) is much more dangerous than hyperglycemia episodes (when the glycemia is above this safety range). Furthermore, the insulin infusion u is considered as the control variable, while the carbohydrate (CHO) in meals r is considered as a disturbance. This disturbance, however, is neither a completely known nor a completely unknown disturbance. In fact, in this work we assume that disturbance is unknown, which avoids the alternative to predict or anticipate the disturbance effect. A completely different problem arises - mainly in the context of predictive control - if disturbances (meals) are assumed to be (even partially) anticipated by means of a signal entering the controller. This issue is not in the scope of this paper, but a perspective for future research.

B. Discrete scheme

Zero older hold (ZOH) discretization method is a standard using in ZMPC to obtain the discrete system [13]. Starting with the continuous model 5:

$$\dot{x}(t) = Ax(t) + Bu(t) \tag{5}$$

$u(t)$ is taken as a discrete input such that:

$$u(t) = u(kT_s), \quad kT_s \leqslant t < kT_s + T_s \tag{6}$$

where (kT_s) corresponds to the present sampling instant and $(kT_s + T_s)$ or $(k+1)T_s$ is the next sampling instant. rewritten the equation (5) and if both sides are multiplied by an integrating factor e^{-At} that's:

$$e^{-At}\dot{x}(t) - e^{-At}Ax(t) = \frac{d}{dt}(e^{-At}x(t)) = e^{-At}Bu(t) \quad (7)$$

which may be integrated directly to give

$$\int_0^t \frac{d}{d\tau}(e^{-A\tau}x(\tau))d\tau = e^{-At}x(t) - x(0) = \int_0^t e^{-A\tau}Bu(\tau)d\tau \quad (8)$$

An analytical solution to the continuous state-space model (5) is given by:

$$x(t_1) = e^{At_1}x(t_0) + \int_{t_0}^{t_1} e^{A(t_1-\tau)}Bu(\tau)d\tau \quad (9)$$

rewritten with the sample time and rearranged to give the state variable response explicitly:

$$x((k+1)T_s) = e^{AT_s}x(kT_s) + \int_{kT_s}^{(k+1)T_s} e^{(k+1)T_s-\tau}bu(\tau)d\tau \quad (10)$$

Then, continuous time solution (9) is mapped exactly to discrete time by ZOH method such that:

$$x(k+1) = A_d x(k) + B_d u(k) \quad (11)$$

Discrete time matrices A_d and B_d are defined as [13]:

$$A_d = e^{AT_s},\ B_d = \int_{kT_s}^{(k+1)T_s} e^{(k+1)T_s-\tau}d\tau = A^{-1}(e^{AT_s} - I)B \quad (12)$$

C. Impulsive scheme

If we assume that the insulin infusion u is injected to the system only at certain time instants given by $\tau_k \triangleq kT$, where T is the fixed period, and $k \in \mathbb{N}$, then it is possible to work under the impulsive system framework.

Formally, assume that the input is given by

$$u(t) = u(kT)\delta(t-kT), \quad t \in [kT, (k+1)T], \quad k \in \mathbb{N}, \quad (13)$$

where $\delta(t)$ is a generalized function (or distribution) Dirac delta[1], that fulfills $\delta(t) = \infty$ for $t = 0$, $\delta(t) = 0$ for $t \neq 0$, and

$$\int_{-\infty}^{\infty} g(\zeta)\delta(\zeta)d\zeta = g(0), \quad (14)$$

for all continuous, compactly supported, function g. This way, the solution of (3) at each periods T can be divided into two parts, the first one, describing the system in the period $[kT,\ kT+\Delta T]$, and the second one describing the system in the period $(kT+\Delta T,\ (k+1)T)$, for a positive and arbitrary small Δ:

$$\begin{aligned}\varphi(t; x(kT), u, r) &= e^{A(t-kT)}x(kT) \quad (15)\\ &+ \int_{kT}^{kT+\Delta T} e^{A(t-\zeta)}B_u u(\zeta)\delta(\zeta - kT)d\zeta\\ &+ \int_{kT}^{kT+\Delta T} e^{A(t-\zeta)}B_r r(\zeta)d\zeta\\ &+ \int_{kT}^{kT+\Delta T} e^{A(t-\zeta)}d\zeta E,\end{aligned}$$

for $t \in [kT,\ kT+\Delta]$, and

$$\begin{aligned}\varphi(t; x(kT+\Delta T), u, r) &= e^{A(t-kT)}x(kT+\Delta T) \quad (16)\\ &+ \int_{kT+\Delta T}^{(k+1)T} e^{A(t-\zeta)}B_r r(\zeta)d\zeta\\ &+ \int_{kT+\Delta T}^{(k+1)T} e^{A(t-\zeta)}d\zeta E,\end{aligned}$$

for $t \in (kT+\Delta,\ (k+1)T)$.

Now, if we consider the limits of this solution for $\Delta \to 0$, it follows that

$$x(kT^+) \triangleq \lim_{\Delta\to 0} \varphi(t; x(kT), u, r) = x(kT) + B_u u(kT), \quad (17)$$

and

$$\begin{aligned}x(t) &\triangleq \lim_{\Delta\to 0} \varphi(t; x(kT^+), u, r) = e^{At}x(kT^+) \quad (18)\\ &+ \int_{kT^+}^{(k+1)T} e^{A(t-\zeta)}B_r r(\zeta)d\zeta + \int_{kT^+}^{(k+1)T} e^{A(t-\zeta)}d\zeta E,\end{aligned}$$

for $t \in (kT^+,\ (k+1)T)$. This latter solution is the one corresponding to the continuous time system $\dot{x}(t) = Ax(t) + B_r r(t) + E$, $x(0) = x_0$. So, the hybrid system evolution can be described as an impulsive system of the form:

$$\begin{cases}\dot{x}(t) &= Ax(t) + B_r r(t) + E, \quad x(0) = x_0, \quad t \neq \tau_k,\\ x(\tau_k^+) &= x(\tau_k) + B_u u(\tau_k), \qquad\qquad\qquad\qquad k \in \mathbb{N},\end{cases} \quad (19)$$

D. Underlying discrete time subsystems

According to the impulsive scheme developed in [14], two underlying discrete-time subsystems associated to (19) can be defined to characterize the states $x(\tau_k)$ and $x(\tau_k^+)$ evolution[2]:

$$\begin{aligned}x^\bullet(k+1) &= A^\bullet x^\bullet(k) + B_u^\bullet u^\bullet(k) + B_r^\bullet r^\bullet(k) + E^\bullet,\ x^\bullet(0) = x(\tau_0),\\ x^\circ(k+1) &= A^\circ x^\circ(k) + B_u^\circ u^\circ(k) + B_r^\circ r^\circ(k) + E^\circ,\ x^\circ(0) = x(\tau_0^+),\end{aligned}$$

where $A^\bullet = A^\circ \triangleq A^d = e^{AT}$, $B_u^\bullet = e^{AT}B_u$, $B_u^\circ = B_u$. Furthermore, the input and disturbance are related by $u^\circ(k+1) = u^\bullet(k) = u(\tau_k)$, $r^\circ(k) = r^\bullet(k) = r(k)$, and $E^\bullet = E^\circ \triangleq \int_0^T e^{A\zeta}d\zeta E$, $B_r^\bullet = B_r^\circ \triangleq \int_0^T e^{A\zeta}d\zeta B_r$.

[1]The Dirac delta is only an abstraction to formulate the impulsive problem. The idea behind this concept is that quick insulin injections can be properly approximated by impulses, if compared to T.

[2]The state $x(\tau_k)$ is denoted as $x^\bullet(k)$, while $x(\tau_k^+)$ is denoted as $x^\circ(k)$.

E. Extended equilibrium for the impulsive representation

The extended equilibrium points of the impulsive representation are characterized by the equilibrium points of the Underlying Subsystems, $(x_s^\bullet, u_s^\bullet, r_s^\bullet)$ and $(x_s^\circ, u_s^\circ, r_s^\circ)$, respectively, in such a way that they must fulfill the conditions:

$$x_s^\bullet = A^\bullet x_s^\bullet + B_u^\bullet u_s^\bullet + B_r^\bullet r_s^\bullet + E^\bullet,$$
$$x_s^\circ = A^\circ x_s^\circ + B_u^\circ u_s^\circ + B_r^\circ r_s^\circ + E^\circ.$$

So, the equilibrium sets corresponding to two subsystems an can be consulted in [14].

III. Zone MPC formulation

In this section is stated the ZMPC formulations for the standard version and the iZMPC developed in [14]. The main objective is to steer the system from a given initial state x0 to an equilibrium objective zone.

A. Standard zone MPC

Main objective of the ZMPC is calculate the optimal future discrete-time inputs according to prediction horizon and a objective cost function to remains the predictive output in the target zone. The control objective is to minimize the cost function of the form [7]:

$$\begin{aligned} \min_{u_k} J &= \sum_{j=1}^{N} \| \left(y(k+j) - y_{ref} - \delta\right) \|_Q^2 \\ &+ \sum_{j=0}^{N} \| \left(u(k+j)\right) \|_R^2 \end{aligned} \tag{20}$$

Subject to restrictions:

$$\begin{aligned} x[k+1] &= A_d x[k] + B_d u[k] \\ u_{min} &\leqslant u[k] \leqslant u_{max} \\ |\delta| &\leq \delta_{max} \end{aligned} \tag{21}$$

Where $y(k+j)$ is the predictive output , δ_{max} it is the maximum change around the set-point, y_{ref}, that generated the control target zone. $u(k+j)$ is the predictive optimal discrete-time input for remains the predictive output in the control objective zone. The matrices Q and R determine the relative importance of error and the cost of energy. The first restriction is the discretization of the model, the second one is the control input bounded in a minimum u_{min} and maximum u_{max} permitted inputs. Finally, the cost function is solved by optimization methods using the tools $Matlab$® y $Model\ Predictive\ Control\ Toolbox$™ .

B. Zone MPC with state window target control

The therapeutic window is characterized - for the impulsive scheme - by sets $\mathcal{X}_s^{\bullet Tar}$ and $\mathcal{U}_s^{\bullet Tar}$ while $\mathcal{X}_s^\bullet$ and $\mathcal{U}_s^\bullet$ denote the equilibrium sets.

The proposed MPC formulation is the one described in [14], adapted to the integrating affine system (3). The cost of the optimization problem that MPC solves on-line is

$$\begin{aligned} V_N(x, \mathbf{r}, \mathcal{X}_s^{\bullet Tar}, \mathcal{U}_s^{\bullet Tar}; \mathbf{u}, u_a, x_a) &\triangleq V_{dyn}(x, \mathbf{r}; \mathbf{u}, u_a, x_a) \\ &+ V_f(\mathcal{X}_s^{\bullet Tar}, \mathcal{U}_s^{\bullet Tar}; u_a, x_a), \end{aligned}$$

where

$$\begin{aligned} V_{dyn}(x, \mathbf{r}; \mathbf{u}, u_a, x_a) &= \sum_{j=0}^{N-1} \left(x(j) - x_a\right)^T C'QC \left(x(j) - x_a\right) \\ &+ \left(u(j) - u_a\right)^T R \left(u(j) - u_a\right), \end{aligned}$$

with $Q > 0$ and $R > 0$, is a term devoted to steer the system to a certain artificial open-loop equilibrium variable given by the artificial pairs $(u_a, x_a) \in \mathcal{U}_s^\bullet \times \mathcal{X}_s^\bullet$, and

$$\begin{aligned} V_f(\mathcal{X}_s^{\bullet Tar}, \mathcal{U}_s^{Tar}; u_a, x_a) &= p\left(dist_{C\mathcal{X}_s^{\bullet Tar}}(Cx_a)\right. \\ &+ \left. dist_{\mathcal{U}_s^{\bullet Tar}}(u_a)\right) \end{aligned}$$

with $p > 0$, is a terminal cost devoted to steer Cx_a to the whole sets $C\mathcal{X}_s^{\bullet Tar}$ and u_a to $\mathcal{U}_s^{\bullet Tar}$, respectively. Matrix C is an output matrix that decides which states will be controlled (we consider in this case the first 3 states, which represent the controllable part of the system).

IV. In silico:Numerical results

The simulation was performed based on the data of three virtual patients taken from [11], called IF2, IF3 and BE The initial condition for x_1 is set up in $250mg/dl$, $128mg/dl$ and $130mg/dl$ respectively. In any condition the first goal is to steer the system from this condition to the normoglycemia zone ($70 < x_1 < 140$). Once the system is in this zone, the second objective is to maintain the system inside these limits even when strong 'perturbations' (as food intake) enter to the system. Since the patients are hospitalized, regular meals are considered: the breakfast at 8 a.m, the launch at 1 p.m. and the dinner at 8 p.m. Furthermore, it is assumed that the counting of CHO in the meals is accurate and known. The parameters of each patient can be consulted in [11].

The standard ZMPC controller is tuned under the following parameters $N = 100$, $R = 1 \times 10^{-8}$, $Q = 3e10$ y $y_{ref} = 100[mg/dl]$ and the state and input constrains are define by $\{[0\ 0\ -0.1\ 0\ -0.1]^T \preceq x \preceq [300\ .5\ 0.1\ 1\ 0.1]^T\}$, $\{0 \leq u \leq 30\}$, respectively.

The iZMPC controller is tuned under the following parameters $N = 52$, $R = 1$, $Q = 500$, $p = 500000$. The state and input constraints are given by $\mathcal{X} = \{x : [0\ 0\ -0.1\ 0\ -0.1]^T \preceq x \preceq [500\ .5\ 0.1\ 1\ 0.1]^T\}$ and $\mathcal{U} = \{u : 0 \leq u \leq 30\}$, respectively. The state target window should be decided by the treating physician. In the simulation below it is defined by $\mathcal{X}^T = \{x : [80\ 0\ -0.1\ 0\ -0.1]^T \preceq x \preceq [100\ 0.2\ 0.05\ 0.5\ 0.05]^T\}$ more details about the tuning can be found in [14]

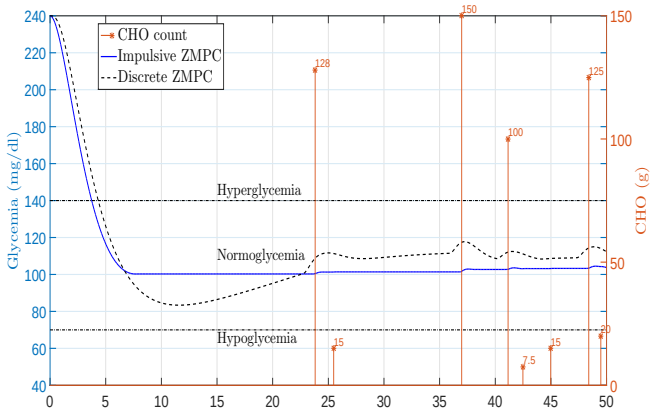

Fig. 1. IF2 patient, Glycemic evolution iZMPC (blue line) and discrete MPC (green line).

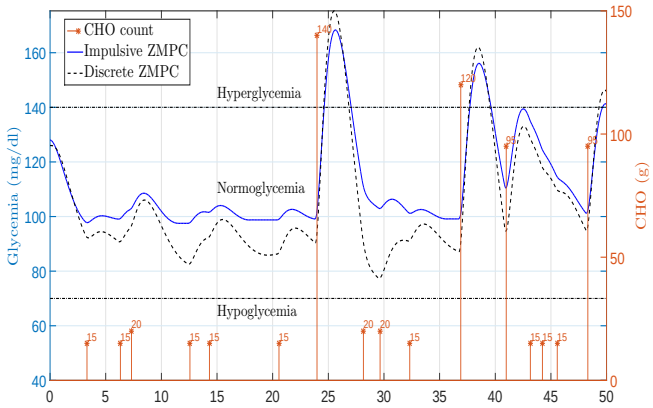

Fig. 2. IF3 patient, Glycemic evolution iZMPC (blue line) and discrete MPC (green line).

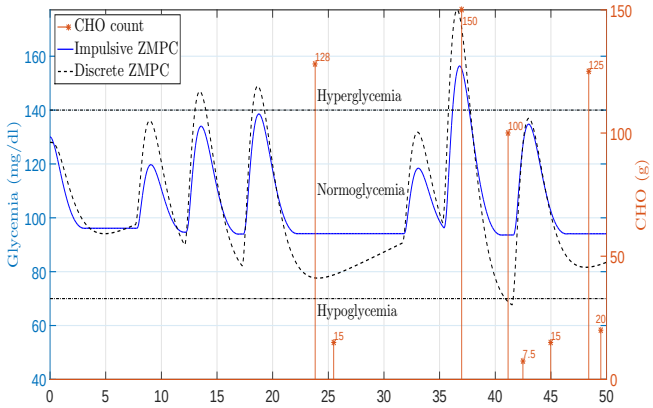

Fig. 3. BE patient, Glycemic evolution iZMPC (blue line) and discrete MPC (green line).

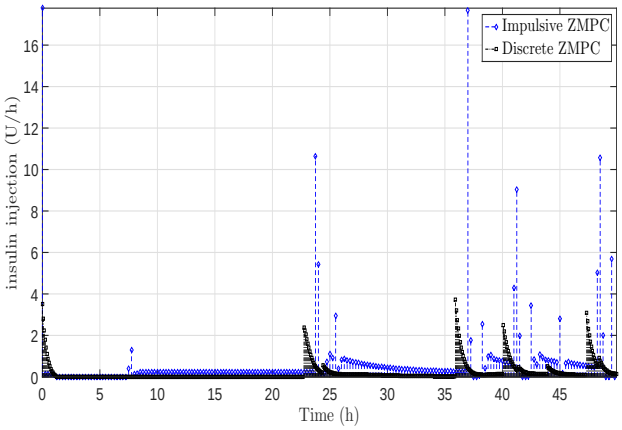

Fig. 4. IF2 patient, insulin infusion iZMPC (black line) and discrete MPC (blue line).

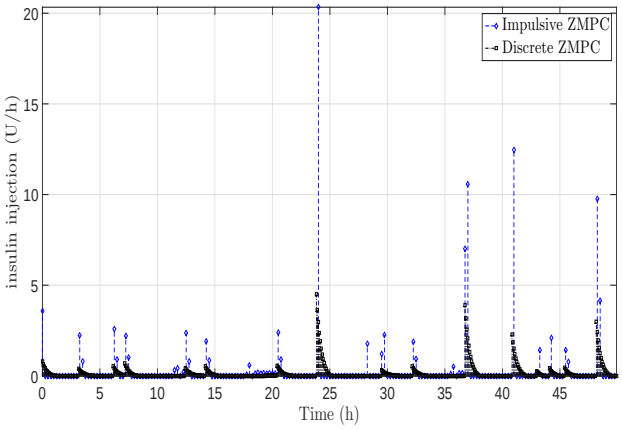

Fig. 5. IF3 patient, insulin infusion iZMPC (black line) and discrete MPC (blue line).

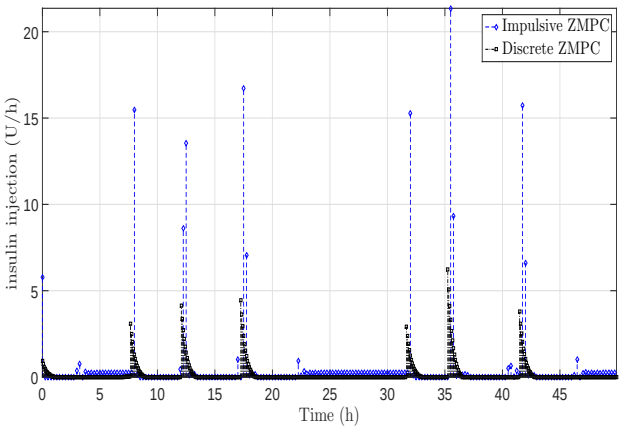

Fig. 6. BE patient, insulin infusion iZMPC (black line) and discrete MPC (blue line).

V. Conclusions

A comparison of the performance between discrete and impulsive form on the design of ZMPC for the regulating glycemia in type I diabetic patients has been tackled in this paper. The main differences between this ZMPC controllers are: i) the iZMPC controller don't need a permanent insulin injection in contrast to the discrete (zero-order-hold) ZMPC, in the iZMPC case the insulin is delivered by boluses, preventing insulin overdoses. ii) both controllers can maintains the glycemia levels in the normal range $70 \sim 140mg = dl$ but the iZMPC obtain a more strong regulation of the glycemic in the normoglycemia zone (see figures 1, 2 and 3).

In general, the main ZMPC parameters may be not trivial, it allows a variety of closed-loop behaviors, and several studies can be done to find a method to tune the controller according to the significant model parameters of the patient. This is necessary to the acceptation by the medical, patients and pumps makers communities. the simulations has good results in this virtual patients, for the future studies its important the validation with a larger cohort of patients.

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